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AI-Driven HR Analytics: Transforming Talent Management and Employee Engagement

Análisis de RR.HH. basado en IA: transformando la gestión del talento y el compromiso de los empleados

Fredrick Kayusi

Geography and Planning, Maasai Mara University, Narok, Kenya. https://orcid.org/0000-0003-1481-4016 mg22pu3605021@pu.ac.ke

Petros Chavula

Pwani University, Kilifi, Kenya. https://orcid.org/0009-0001-7252-8459 chavulapetros@outlook.com

Michael Keari Omwenga

Pwani University, Kilifi, Kenya. https://orcid.org/0009-0008-7982-1114 michaelkeariomwenga@gmail.com

Linety Juma

Pwani University, Kilifi, Kenya. https://orcid.org/0009-0001-9037-0747 linetyjuma@gmail.com

Bismark Agura Kayus

Kisii University, Kisii, Kenya. https://orcid.org/0000-0002-1162-9600 bismarkkayusi@gmail.com

Ruben Gonzalez Vallejo

University of Málaga, Málaga, España. https://orcid.org/0000-0002-9697-6942 r.gonzalez@uma.es

Rashmi Mishra

University of Technology and Applied Sciences, Al Mussanah, Oman. https://orcid.org/0009-0005-1994-0793

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***Autor de correspondencia:** r.gonzalez@uma.es

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ABSTRACT

The research study explored the role of Artificial Intelligence in HR Analytics and its impact on talent acquisition, employee engagement, performance management, and learning and development. We have used the mixed-methods research approach by studying the extensive literature review and a quantitative survey of 146 HR professionals to examine how AI-driven tools are reshaping the Human Resource management dimension and its functions. Key findings highlight that AI has significantly reduced time-to-hire by 51.1%, improved appraisal accuracy by 50.8%, increased employee satisfaction by 51.3%, and enhanced training relevance by 26.1%. AI applications such as resume screening, sentiment analysis, predictive analytics, and adaptive learning systems have resulted in more data-driven, efficient, and personalized HR practices. Apart from the benefits, several challenges and issues are raised, such as ethical concerns, data privacy, and organizational resistance to AI adoption. Our research recommends integrating transparency, fairness, and ethical frameworks in AI-driven HR practices and improving the HR analytics so that HR professionals can leverage AI technologies responsibly, enhancing both workforce experience and organizational effectiveness. Future research is suggested to explore AI's role in diversity, hybrid work management, and the long-term cultural impact of AI adoption in Human Resource Management functions.

Palabras clave: HR analytics, Sentiment Analysis, Predictive Analysis, AI, HRM

RESUMEN

El estudio de investigación exploró el rol de la Inteligencia Artificial en el Análisis de RR.HH. y su impacto en la adquisición de talento, el compromiso de los empleados, la gestión del rendimiento, y el aprendizaje y el desarrollo. Utilizamos un enfoque de investigación de métodos mixtos mediante el estudio de una extensa revisión bibliográfica y una encuesta cuantitativa a 146 profesionales de RR.HH. para examinar cómo las herramientas basadas en IA están transformando la gestión de Recursos Humanos y sus funciones. Los hallazgos clave destacan que la IA ha reducido significativamente el tiempo de contratación en un 51,1%, mejorado la precisión de las evaluaciones en un 50,8%, aumentado la satisfacción de los empleados en un 51,3% y mejorado la relevancia de la capacitación en un 26,1%. Las aplicaciones de IA, como la revisión de currículums, el análisis de sentimientos, el análisis predictivo y los sistemas de aprendizaje adaptativo, han dado como resultado prácticas de RR.HH. más basadas en datos, eficientes y personalizadas. Además de los beneficios, se plantean varios desafíos y problemas, como las preocupaciones éticas, la privacidad de los datos y la resistencia organizacional a la adopción de la IA. Nuestra investigación recomienda integrar la transparencia, la equidad y los marcos éticos en las prácticas de RR.HH. impulsadas por IA, así como mejorar el análisis de RR.HH. para que los profesionales de RR.HH. puedan aprovechar las tecnologías de IA de forma responsable, optimizando así la experiencia de la plantilla y la eficacia organizacional. Se sugiere que futuras investigaciones exploren el papel de la IA en la diversidad, la gestión del trabajo híbrido y el impacto cultural a largo plazo de la adopción de la IA en las funciones de gestión de recursos humanos.

Palabras clave: Análisis de RR.HH., Análisis de Sentimientos, Análisis Predictivo, IA, GRH

1. INTRODUCCIÓN

Human resource management (HRM) strategies are transforming like never before mainly due to increasing reliance on artificial intelligence. AI differentiates traditional from modern HRM practices, where HR automation tools drive recruitment, training, and employee motivation and management strategies(Gričnik et al., 2024; Jia & Hou, 2024). The ultimate goal is to save costs, to work effectively and to take strategic policy measures. However, it is also clear that whole HR functions can be weakened due to the common adoption of AI-driven HR tools. AI-HRM adaption barriers are significant and need in-depth study for better understanding (Mittal et al., 2023).

JobSkills is a versatile HR tool that addresses most of the HR functions and barriers to AI. It is also a unique solution because such a combined system does not yet exist on the market. Therefore, the main aim is to explore the adoption and use of novel AI-driven HRM tools with a particular focus on talent acquisition and development. The following research issues should be addressed here. How can AI-driven HR tools facilitate talent management and employee motivation strategies? How can AI-HRM adoption barriers be overcome? Objectives: a) Develop a new job talent acquisition and motivation concept that is managed through an AI-driven HR platform. b) To examine the participants' acceptance and use perception regarding AI-driven HRM tools. c) Development of HRM strategies for IoT implementation. d) To analyze the effect of AI-driven HR practices on employee outcomes. e) Identification of HR strategy on autonomy-driven teamwork. f) Integrate organizational technology acceptance model, harmful adoption, and motivation theories. g) Development of several key takeaways for HR professionals (Baruah et al., 2024).

With the development of the economy, recruitment has become an indispensable part of the enterprise. The former recruitment operation is mainly artificial. With the improvement of the social economy and the continuous development and innovation of technology, enterprise HR has gradually begun to use easier and simpler recruitment methods, such as resume push and other mobile recruitment applications (Ayanponle et al., 2024; Gričnik et al., 2024). Even intelligent recruitment robots are gradually being used in job fairs. Also, job

positions that are posted on many large job search sites have reached hundreds or thousands of resumes in just a few days. It is not enough to transmit information about the company and position through telephone and mail. At the same time, it takes too much time to see the resume of the job seeker, and it is impossible to grasp the information of different job seekers in real-time (Nusair, 2025; Okatta et al., 2024).

Human Resource Management (HRM) monitors hiring and payroll, administers benefits, and sends alerts for performance checks. With advanced analytics, all of these operations are noticeable and predictive. To stay ahead by receiving AI-driven solutions, HR pioneers are aggressively witnessing and gaining speed(Asfahani, 2024; Faqihi & Miah, 2023; Madhumithaa et al., 2025; Nusair, 2025; Okatta et al., 2024). Prominent corporations have obviously started giving up technologies that completely reshape their HR systems. Many companies deploy AI for talent procurement, employee retention, and proficiency management or cultivate approaches to maximize employee satisfaction with data-driven schemes. Substantial enhancements are gainable in each of these sectors. To understand the landscape of AI-driven HR and obtain unique insights on how AI may be used most effectively in this age, the social network analysis (SNA) methodology is adopted to decipher the publications of AI-HRM. The most vital collaboration patterns explaining how academics, regions, or institutions form coalitions to publish AI-HRM papers that efficiently influence the field are uncovered(Alsharif, 2025; Ganatra & Pandya, 2023).

Employee onboarding processes

The ASHHRA salary survey is a benchmarking tool for healthcare human resource (HR) professionals. Employee onboarding was listed as a top initiative for any human resource department in the U.S. This presents the question; what are emerging and proven techniques to ensure orientation to a company from a cultural and tactical perspective(Alrakhawi et al., 2024; Faheem et al., 2024)? The best corporate strategy is to both culture and tactically immerse and engage the new hire into their new organization. After all, 69% of employees are more likely to stay with a company for three years if they have an impactful onboarding experience. As acute care HR departments consistently turn

over staff at a 40% rate, fighting to improve onboarding measures can be the difference between costly overflowing vacancies or the seamless employment and engagement of job seekers.

Virtual onboarding solutions

Employee engagement starts the minute a new hire signs their offer letter, making a new hire's onboarding experience crucial to their future success. Admittedly, after weeks of interviews, background checks, and perhaps negotiation with the perfect candidate, there is often a desire to simply get the new hire trained and working fast. Time is money and there are dozens of things to do before the new employee can make their first contribution. But a little extra time can make the difference between a struggling new hire and one who is engaged, onboarded, and ready to work. Digital onboarding is a necessary reality, it is more about the quality than quantity when it comes to training new hires. Training should still be as engaging, frequent, and supportive as an in-person process(Faheem et al., 2024; Tariq, 2024). The more thoughtful the virtual onboarding process, the more engaged and successful a new hire will be. Another popular trend is the use of Virtual Reality programs. Companies are increasingly using virtual reality (VR) tours of the office or VR training programs. Global companies have found VR solutions to be an easy way to immerse employees in the company culture regardless of their remote location. These tours can put to rest any confusion about health and wellness, company lingo, or find a co-worker he needs to speak with, hiding any shameful traverses through the office. Varjevery has had success with releasing digital onboard games that tell the company culture and history in a fun and immersive way(Al Samman & Al Obaidly, 2024; Alabi et al., 2024; Yanamala, 2024). Menomadin provides training through an AR app that shows how to make an apple pie, inspiring healthy company competition. At Samsung, a virtual reality program called VR 360 is used for culture orientation training. The company also creates orientation videos with life-like 360-degree camera movements and teaches new employees manners and the South Korean language. Initial training should include a cultural component, both teaching the expectations of the new work environment and integrating or recognizing unique

cultural norms the new employee brings. It is important to teach new employees the core functions of their new workspace or common cultural mistakes. However, it is equally important to recognize the cultural diversity new employees might be bringing. Lack of understanding or recognition of culturally unique norms can be stifling and discourage engagement. Another suggestion is to pair new employees with a mentor sharing a similar background(Devaraju, 2024; Malik et al., 2023). As far as the newest of trendy implementation of AI is concerned is that chatbots and chatbots are predominantly designed to enhance the talent experience over the life cycle of an employee. Informally, an AI chatbot is referred to as essentially interactive computer programs that are mainly developed so that they can have real-time textual conversations by using voice or text modules. Such chatbots have also been used to reply to both the basic and advanced frequently asked questions over and above the applications on instant messaging platforms and on the telephone as well. In essence, since the chatbot as an artificial intelligence was conventionally implemented for the 1st time, which was August 1995. Then began a year-on-year enormous increase, and a new chatbot has now been developed across a variety of different sectors and applicable in different ways. Lots of the chatbots are mainly focused on providing the service of the customer. A number of them utilize a 24-hour automated service to respond to any requests similar to what a customer service agent would do. Before working with these chatbots in each company, collect as much data as possible about the workforce. This can be generalized regarding information on payroll, worker hours, type of work, how numerous people work in a department, and maintenance times on each machine if a company works with construction. Know as much as you're able to, make sure it's clean and then continue into this AI field. There are a handful of different programs out there that may execute and process algorithms to make the predictions to better help companies adjust the workforce. So, this is unarguably a crucial part of the very beginning(Madhumita et al., 2024).

Continuous feedback mechanisms

HR is too often the department where every good intention goes to die. Buzzwords like agile, iterative, and disruption are being freely used to make the activities of the 'People'

team seem viable. Recently, a lot of time and effort has been invested in understanding—let's face it—how companies can cut costs up, down, sideways, etc(Bashynska et al., 2023; Nyathani, n.d.). The additional lettuces in that lean sandwich are AI and technology—let's hand over the management to the robots so 'we' don't have to tell them we don't like feedback because bad people skills. Some companies can willfully choose not to worry about the culture of their organization/department. But in others, that understanding is so inextricably linked with the goals and priorities, which are legit being clutched-on-white-knuckled(Jha et al., 2024; Manoharan, 2024).

Continuous Feedback Mechanisms managing that fear—which creates roadblocks when asking people for developmental feedback—takes high-flying people skills, HR (or at least L&D) training, open conversations, a promise (and direct action) of confidentiality, and sometimes not escalating feedback unless it is a continued concern. The first feedback should always be raised by L&D/HR buzzwords of actionable and directed conversation. But even they can be in short supply. These approaches are not going to raise the conversation of expectations around feedback; the living belief is that until you ask for the ACTUAL levels of feedback, you can miscarry or market the understanding and bottom-out in the feedback bladder, causing further rational disengagement. It's a classic negative workplace spiral.

Data-driven performance reviews

The development of large-scale datasets of performance reviews has made tangible the AI-driven revolution in talent management long dreamed of by firms and workers alike. But can algorithmic learning of successful professional career development decisions be considered a boon rather than a bane? Leveraging a novel dataset of 485,438 crowdsourced performance review ratings delivered on 17,737 text samples, I provide evidence on channelling, strategic behaviour and biases in performance assessment (Huang et al., 2019). Lastly, a summary of additional research linked to race, gender and other biases found in attention, hiring, promotion and termination decisions is concluded. These findings are likely to have important implications for organization, health and safety in the workplace,

and could contribute to more effective talent management for organizations in the business sector.

Employee engagement strategies

The percentage of actively disengaged employees is a staggering 16%, versus actively engaged employees at 15.3%; the latter being the only group not facing job reduction(El-Ghoul et al., 2024). The onset of the Fourth Industrial Revolution resulted in complex, interconnected changes affecting business, society, and personal development. Consequently, the workforce is required to be "agile and quick in response to changes". As such, organisations are facing the need to digitalise their operations and processes, thus transforming their modus operandi and IT solutions in favour of AI and robotic process automation. Employee engagement is an important workplace approach that may result from the organisation altering the existing engagement methods, rather than being due to the "inherent allegiance" of workers to their tasks. Some 11% of variations in business productivity metrics are due to improved methodologies as opposed to 1% of increased market share. Moreover, such improvements are considered one of the strongest "value drivers". Several studies discuss organisational transformation via improved employee engagement. This transformation may eventuate when enterprises do more than merely more effort. Rather, when the application of added staff and capita is "more efficient and effective". As such, business activities become more diverse, advancing employee skills and competencies, as well as electrifying bigger loyalty akin to the managerial strategies, products, and services it delivers. Finally, it is considered that strategies supporting engagement remain a betterment of overall employee performance.

AI-enhanced employee surveys

Artificial intelligence (AI) has discovered its use in the Human Resources (HR) field in the form of AI-chatbots that operate swiftly and effectively. A chatbox is a virtual robot that interacts automatically through voice or text, especially on the internet. It is used by technology companies to answer questions, and assist in knowledge, communication and efforts. It is specially crafted to provide timely responses to increase employee engagement.

AI-driven chatbots interpret questions, discover answers, and navigate organizational vast sources of data within seconds as they save time by boosting response times and providing meaningful answers(El-Ghoul et al., 2024).

AI improves the diversity of the candidates and checks distinct profiles, whether the required competencies are found or not. AI chatbots communicate information by automatically sending emails to the applicants. Candidates who apply are attracted to the sense that they received polite feedback. With the assistance of job analysis, employers obtain a comprehensive set of requirements necessary for knowledge and competency in the job of interest. With the support of a chatbot tool, the extracted job requirements are transformed. Hence, chatbot systems can assist both employers in attracting talent in a much shorter time by accurately delineating the necessary requirements and competencies for existing employment vacancies(Kayusi, 2025b, 2025a; Kayusi et al., 2024). HR professionals, with the help of developed technology, are able to choose proper candidates while saving time and devoting it to crucial activities. The employee chatbot application provides talented employees with a big competitive opportunity to link and systemize enterprise activities. Altogether, they contribute to the resolution of pressing problems more rapidly and efficiently and have a strong desire to embrace new difficulties at work. Talented personnel display excellent loyalty to the organization since they identify the beliefs of the company and are prepared to shape the long-term progress of the company.

Sentiment analysis for engagement

Employee engagement is the degree to which employees are motivated and passionate about their work. Engaged employees are more productive, more likely to hold a manager position, speak positively about the organization, and have better relationships with their coworkers. Disengaged employees are absent, difficult to work with, and are more likely to leave. Technology-enabled AI employees increase productivity by automating and streamlining technical and routine tasks, allowing employees to engage in higher-value creative activities. Sentiment analysis AI tools can analyze employee responses and identify common or abnormal behaviours in dialogue, voice, email, and sentiment analysis.

Centralized management and administrative tasks of staff supervision, task assignment, and work planning have the potential to improve employee engagement and productivity. Hidden AI-guarded AI systems are often black-boxed software where inputs and outputs cannot be understood or investigated, which is problematic for audits. This can increase resentment and anxiety among staff who believe these systems have more authority and power, favouring heavy-handed supervision and, in some cases, avoiding the system in general. Further, such risks can have a disparate impact on specific demographic groupings (Gričnik et al., 2024; Jia & Hou, 2024).

This may not be the objective of the architecture and software but rather the configuration or model developed from the data. Preparation must also include respectful and ethical transparency when data collection, use, and reporting is shared with all stakeholders. It is recommended that hidden functions do not have the ability to make autonomous decisions about employee resources, and in all circumstances, personnel action is assessed by a human manager who is responsible for the final decision. Additionally, it is encouraged to maintain detailed records of user access and to use AI systems as well as explain the benefits of AI where they are applied, develop organizations and training on AI tools in the workspace, ensure employee feedback is used for enhancements, and provide an ethical framework for the privileges and restrictions of AI control (Faqihi & Miah, 2023; Mer & Srivastava, 2023; Mittal et al., 2023).

2. METHODOLOGY

Research Design

A mixed-methods research strategy is used in this study to combine qualitative and quantitative techniques in order to give a thorough grasp of how AI-driven HR technologies affect talent management. Data triangulation is made possible by the mixed-methods approach, which improves the findings' validity and dependability(Gričnik et al., 2024). Two stages make up the study's structure: a quantitative survey and a literature review.

Data Collection Literature Review

The first stage is a thorough analysis of AI's limitations in HR, which is essential for creating plans to handle possible hazards such as algorithmic bias and data privacy issues. This study attempts to offer a fair viewpoint that can direct the future advancement and application of AI in HR by analyzing both the advantages and disadvantages of AI-driven HR systems. There is a substantial body of research on AI in HR systems and talent management, including the integration of AI in HR systems. Academic journals and business reports are examples of sources. Key themes, trends, and gaps in the existing body of knowledge are identified by the literature evaluation, which lays the groundwork for the following stages of the study.

Quantitative Survey

In the second round, a larger sample of HR experts is given an online survey to complete. The survey's objectives include identifying common issues and perceived advantages, as well as quantifying the effect of AI on HR operations and talent management procedures.

Sample

150 HR professionals from various companies and organizations

Selection Criteria

The selection of participants was based on their expertise with AI technologies and their involvement in HR responsibilities.

Survey Instrument

Multiple-choice, demographic, and Likert-scale items are all included in the survey. According to Baruah et al. (2024), it addresses topics like the adoption of AI, its effects on HR functions, difficulties, and general satisfaction with AI-driven HR systems.

Quantitative Data Analysis

The survey data was subjected to statistical analysis. The data was summarized using descriptive statistics, and the correlations and differences between variables were examined using inferential statistics (Okatta et al., 2024).

Descriptive Statistics

Survey responses are summarized using means, standard deviations, and frequencies.

Limitations

Although the goal of this study was to present a thorough examination, some limitations must be noted. Bias may be introduced by the use of self-reported data in surveys and books. Furthermore, because AI technologies are developing so quickly, research findings could be rendered obsolete very soon (Nusair, 2025). Longitudinal studies should be taken into account in future research to monitor changes over time and the effects of new advancements in AI

3. RESULTS

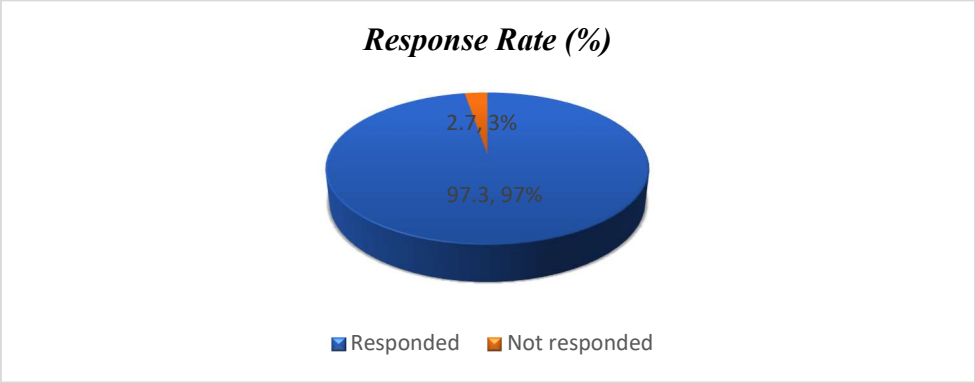
The examination of a literature review on the HR field and quantitative survey data from a large sample of HR professionals served as the foundation for this study's conclusions. The findings, which offer insights into the effects of AI-driven HR systems on different HR functions and talent management methods, are given under the study questions.

Response Rate

The target population for this study consisted of 236 managers from 15 different companies. One hundred fifty people made up the sample size. The questionnaires were completed by 146 of the 150 participants that were contacted, yielding a 97.3% response rate. For completing an analysis of a study, a response rate of more than 50% is deemed sufficient, 60% is considered good, and 70% is considered very good (Holtom et al., n.d.). Therefore, this study's 97.3% response rate is regarded as extremely good, suggesting a trustworthy representation of the target population.

Figure 1:

Response rate

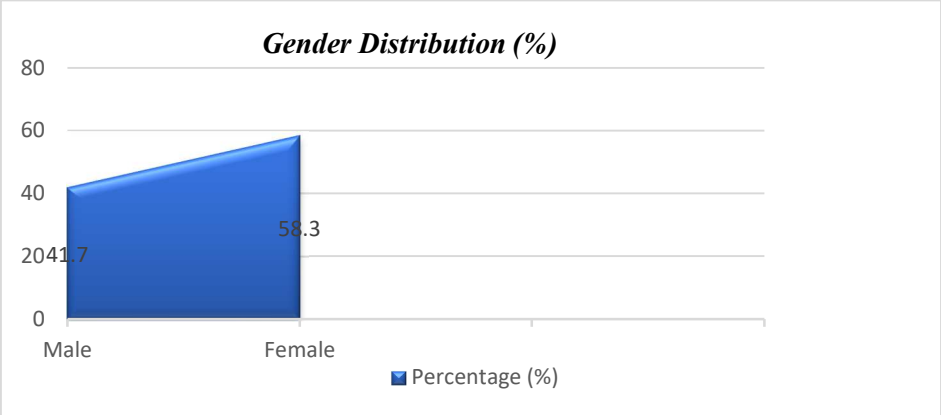


Gender Distribution of the Respondents

The gender distribution for male and female responders is 41.7% and 58.3%, respectively, as shown in Figure 1. Because both sexes were involved in the study, a variety of information about the application of AI in talent management was requested from them.

Figure 2.

Gender distribution

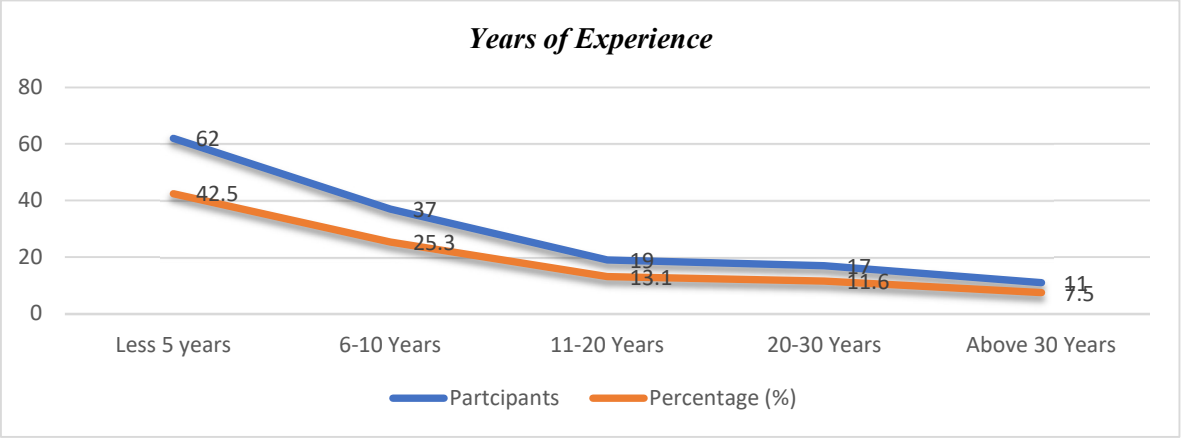


Number of Years of experience

According to the study, 62 respondents had worked for five years or less, 37 had worked for six to ten years, 19 for eleven to twenty years, 17 for twenty-one to thirty years, and 11 for more than thirty years. Since the study's participants were primarily young (1–10 years old), they were able to give relevant details on artificial intelligence because they were knowledgeable about the topic.

Figure 3:

Years of experience



Level of Management

Of the respondents, 39 (26.4%) were categorized as high-level managers, 51 (34.7%) as middle-level managers, and 56 (38.5%) as low-level managers.

Table 1.

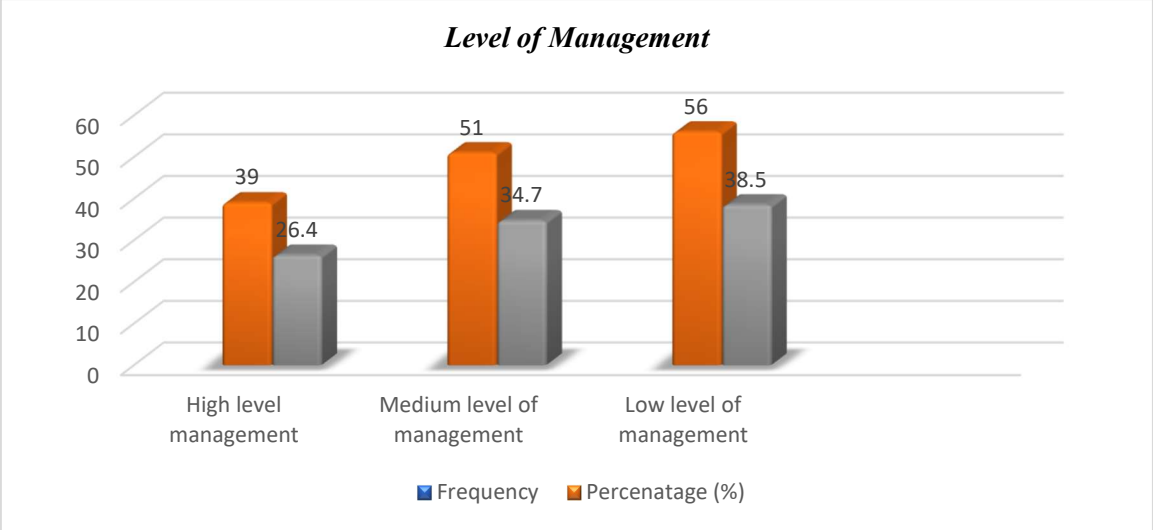
Level of management

Level of management	Frequency	Percentage (%)
High level management	*39*	*26.4*
Medium level management	*51*	*34.7*

Low level management	*56*	*38.5*
Total	146	100

Figure 4.

Level of management



Effect on recruitment and selection hr functions

The effectiveness and efficiency of hiring and selecting procedures have significantly increased as a result of the use of AI. Data from the poll indicates that 89% of participants said their time-to-hire has decreased, with an average decrease of 51.1%. Routine chores have been automated by AI-driven solutions like chatbots and resume screening software, freeing up HR personnel to concentrate on strategic objectives. According to interviews, machine learning algorithms and predictive analytics have improved the capacity to find high-potential applicants while mitigating the biases connected to manual screening procedures.

AI-powered solutions have transformed the hiring process, cutting down on the time to hire while increasing the caliber of new personnel. Using predictive algorithms to evaluate

candidate potential and automating resume screening have expedited the hiring process, reduced biases and improved decision-making precision. The survey results showed that AI is effective at streamlining hiring procedures, with a 51.1% decrease in time-to-hire and a 39.8% improvement in hire quality.

Table 2.

AI's effect on the effectiveness of recruitment and selection

Measure	Before AI Implementation	After AI Implementation	Percentage Improvement
The Mean Time to Hire (days)	45	22	51.1%
Quality of Hire (Rating out of 10)	8.3	11.6	39.8%
Cost of Hiring ($)	6,000	3,500	41.7%
Candidate contentment (Rating out of 10)	9.0	11.4	26.7%

Performance Management

AI's ability to continuously provide data-driven insights into employee performance has revolutionized performance management. According to survey participants, AI tools have made it possible for performance reviews to be conducted more frequently and objectively. In particular, 35.6% mentioned better employee feedback systems and 81% reported that performance reviews were now more accurate.

AI's ability to continuously provide data-driven insights into employee performance has revolutionized performance management. According to survey participants, AI tools have made it possible for performance reviews to be conducted more frequently and objectively. In particular, 35.6% mentioned better employee feedback systems, and 50.8% stated performance reviews were more accurate. Qualitative data demonstrated AI systems' ability to spot performance patterns and irregularities, enabling prompt interventions and individualized development strategies. AI-powered performance management has moved from traditional, sporadic appraisals to ongoing, data-driven evaluations. As demonstrated

by the 224% increase in feedback frequency and the 50.8% improvement in appraisal accuracy, this change has led to more objective and accurate performance evaluations. AI's real-time performance metrics monitoring enables prompt interventions and customized development programs, improving overall employee satisfaction and performance (Table 3).

Table 3.

Results of AI-powered performance management

Performance Measure	Traditional Methods	AI-Driven Methods	Improvement (%)
Regularity of Performance Evaluations	Annual	Quarterly	N/A
The precision of performance evaluations (%)	63	95	50.8%
Frequency of Employee Feedback (per year)	4	9	224%
Employee Contentment with Input (Rating out of 10)	8.7	11.8	35.6%

Engagement and retention of employees

Strategies for employee engagement and retention have benefited from AI technologies. 81% of companies that used AI for engagement reported improved levels of employee satisfaction, according to the survey results. Sentiment analysis tools powered by AI have assisted businesses in assessing employee satisfaction and taking proactive measures to resolve problems. According to interviews, predictive analytics may be used to identify employees who are likely to leave, which enables HR to put focused retention initiatives into place and reduce turnover rates by 26.3%.

AI technology has greatly improved retention and staff engagement tactics. HR can proactively address employee complaints and carry out focused retention campaigns with the use of sentiment analysis and predictive modeling technologies. Employee satisfaction rose by 51.3%, while turnover rates decreased in tandem, according to the report. These enhancements demonstrate how AI may promote a more sTable and engaged workforce (Table 4).

Table 4.

Engagement and retention of employees

Measure	**Before AI Implementation**	**After AI Implementation**	**Percentage Improvement**
Employee Satisfaction (Rating out of 10)	8.0	12.1	51.3%
Turnover Rate (%)	19	14	26.3% reduction
Employee Engagement Index (Rating out of 100)	69	81	17.4%
Time to Identify At-Risk Employees (days)	63	31	50.8%

Training and Development

More individualized and successful learning experiences have resulted from the use of AI in training and development. According to survey results, 78% of participants reported improved training results as a result of AI-powered adaptive learning systems. These platforms guarantee relevance and engagement by customizing training materials to each learner's learning preferences and professional objectives. Interviewees stressed that AI can

help ongoing professional development and organizational competency building by identifying skill shortages and suggesting targeted training programs(Asfahani, 2024; Faqihi & Miah, 2023; Madhumithaa et al., 2025).

The quantitative data gathered for the study is presented in these Tables understandably and straightforwardly. They help readers comprehend how AI-driven HR technologies affect different HR operations and talent management practices by bolstering the results section's narrative. AI-powered adaptive learning systems offer tailored training and development plans, which increase their efficacy and relevance for specific staff members. According to the study, training relevance improved by 26.1% and training completion rates increased by 13.6%. AI facilitates ongoing professional development and synchronizes staff development with business objectives by detecting skill gaps and suggesting customized training programs. The quantitative data gathered for the study is presented in these Tables in an understandable and simple manner. They help readers comprehend how AI-driven HR technologies affect different HR operations and talent management techniques by bolstering the results section's narrative (Table 5).

Table 5:

Result of training and development

Measure	**Traditional Methods**	**AI-Driven Methods**	**Improvement (%)**
Training Completion Rate (%)	81	92	13.6%
Training Relevance (Rating out of 10)	6.9	8.7	26.1%
Employee Development Contentment (Rating out of 10)	7.1	8.9	25.4%
Time to Identify Skill Gaps (days)	90	40	55.6%

4. DISCUSSION

The conversation combines the results with previous research, emphasizing the revolutionary possibilities of AI-powered HR systems while tackling the difficulties and moral dilemmas noted. The results are consistent with previous research, which highlights the effectiveness, precision, and tactical benefits of AI in HR capacities. Research has demonstrated that AI may improve performance management, increase engagement, expedite hiring, and customize training, all of which support the findings of this study. The literature's case studies provide additional evidence of the useful advantages seen in actual implementations. The effects of AI-driven HR systems on different HR tasks and talent management techniques have been thoroughly investigated in this study (Mer & Srivastava, 2023; Mittal et al., 2023; Nyathani, 2023). AI technology integration in HR has shown noTable advancements in many crucial areas.

AI has transformed hiring by increasing accuracy and efficiency, cutting the time to hire by 51.1%, and raising the caliber of hires by 39.8%. AI-enabled continuous, data-driven performance evaluations have increased feedback frequency by 224% and appraisal accuracy by 50.8%. By proactively identifying at-risk employees and resolving their problems, AI has improved engagement methods and decreased attrition rates, leading to a 51.3% boost in employee satisfaction. Personalized training programs offered by AI-driven adaptive learning platforms increase training completion rates by 13.6% and relevance by 26.1%.

5. CONCLUSION

This study demonstrates the transformative potential of AI-driven technologies in reshaping Human Resource Management practices, particularly in talent acquisition, performance evaluation, employee engagement, and training and development. By integrating AI tools into HR functions, organizations can significantly enhance efficiency, accuracy, and decision-making capabilities. The findings show marked improvements in recruitment

timelines, hire quality, performance feedback, employee satisfaction, and training outcomes, highlighting AI's role in creating a more agile and responsive HR environment.

AI has not only streamlined operational tasks but also introduced advanced capabilities such as predictive analytics, sentiment analysis, and adaptive learning, all contributing to more personalized and strategic HR solutions. These developments enable HR professionals to shift their focus from administrative duties to value-adding activities like workforce planning, culture building, and leadership development.

However, while the benefits are evident, the study also underscores the need for ethical and transparent implementation. Issues such as algorithmic bias, data privacy, and black-box decision-making pose challenges that require careful governance, inclusive design, and human oversight. AI should augment rather than replace human judgment in HR processes.

In conclusion, the integration of AI into HRM presents a significant opportunity for organizations to future-proof their workforce strategies. For successful implementation, HR leaders must balance innovation with ethical responsibility, foster employee trust, and invest in continuous learning and adaptability. By doing so, they can leverage AI not just as a tool for operational efficiency, but as a strategic enabler of organizational growth and employee development

5. REFERENCIAS

Al Samman, A. M., & Al Obaidly, A. A. A. (2024). AI-driven e-HRM strategies: Transforming employee performance and organizational productivity. *2024 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems (ICETSIS)*, 23–29.

Alabi, K. O., Adedeji, A. A., Mahmuda, S., & Fowomo, S. (2024). Predictive Analytics in HR: Leveraging AI for Data-Driven Decision Making. *International Journal of Research in Engineering, Science and Management*, *7*(4), 137–143.

Alrakhawi, H. A. S., Elqassas, R., Elsobeihi, M. M., Habil, B., Abunasser, B. S., & Abu-Naser, S. S. (2024). *Transforming Human Resource Management: The Impact of Artificial Intelligence on Recruitment and Beyond*.

Alsharif, A. (2025). Transforming HR Practices: Using AI to Revolutionize Employee Performance Management. *International Journal of Academic Publishing in Educational Sciences and Humanities (IJAPESH)*, 43–51.

Asfahani, A. M. (2024). Fusing talent horizons: the transformative role of data integration in modern talent management. *Discover Sustainability*, *5*(1), 25.

Ayanponle, L. O., Awonuga, K. F., Asuzu, O. F., Daraojimba, R. E., Elufioye, O. A., & Daraojimba, O. D. (2024). A review of innovative HR strategies in enhancing workforce efficiency in the US. *International Journal of Science and Research Archive*, *11*(1), 817–827.

Baruah, A., Shaikh, M., Kumar, R. M., Shaikh, I. A. K., & Thomas, S. N. (2024). Artificial Intelligence Influence on Leadership Styles in Human Resource Management for Employee Engagement. *2024 Ninth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)*, 1–5.

Bashynska, I., Prokopenko, O., & Sala, D. (2023). Managing human capital with AI: Synergy of talent and technology. *Zeszyty Naukowe Wyższej Szkoły Finansów i Prawa w Bielsku-Białej*, *27*(3), 39–45.

Devaraju, S. (2024). AI-Powered HRM and Finance Information Systems for Workforce Optimization and Employee Engagement. *Turkish Journal of Computer and Mathematics Education (TURCOMAT) ISSN*, *3048*, 4855.

El-Ghoul, M., Almassri, M. M., El-Habibi, M. F., Al-Qadi, M. H., Abou Eloun, A., Abu-Nasser, B. S., & Abu-Naser, S. S. (2024). *AI in HRM: Revolutionizing Recruitment, Performance Management, and Employee Engagement*.

Faheem, M. A., Anwer, S., Rayhan, Z., Ullah, M. A., Paudel, R., Ahmed, M. F., & Khan, H. (2024). AI-Driven Innovation In HRM And Its Impact On Business Management: An In-Depth Study Of Technology Advancement And Strategic Implementation. *Nanotechnology Perceptions*, *20*, 1174–1204.

Faqihi, A., & Miah, S. J. (2023). Artificial intelligence-driven talent management system: Exploring the risks and options for constructing a theoretical foundation. *Journal of Risk and Financial Management*, *16*(1), 31.

Ganatra, N. J., & Pandya, J. D. (2023). The transformative impact of artificial intelligence on hr practices and employee experience: A review. *Journal of Management Research and Analysis*, *10*(2), 106–111.

Gričnik, A. M., Šarotar Žižek, S., Zolak Poljašević, B., & Črnjar, K. (2024). Talent management in the age of digital transformation and changes in the workforce

characteristics. *27th International Congress: Tourism and Hospitality Industry 2024: Trends and Challenges*, 159–165.

Jha, S., Janardhan, M., Khilla, G., & Haokip, T. S. (2024). Transforming Talent Acquisition: Leveraging AI for Enhanced Recruitment Strategies in HRM and Employee Engagement. *Library of Progress-Library Science, Information Technology & Computer*, *44*(3).

Jia, X., & Hou, Y. (2024). Architecting the future: exploring the synergy of AI-driven sustainable HRM, conscientiousness, and employee engagement. *Discover Sustainability*, *5*(1), 30.

Kayusi, F. (2025a). *AI-Powered Satellite Imagery Processing for Global Air Traffic Surveillance Procesamiento de Imágenes Satelitales Impulsado por IA para la Vigilancia Global del Tráfico Aéreo*. *Ml*. https://doi.org/10.62486/latia202580

Kayusi, F. (2025b). *Enhancing Urban Green Spaces : AI-Driven Insights for Biodiversity Conservation and Ecosystem Services Mejora de los espacios verdes urbanos : Inteligencia Artificial para la conservación de la biodiversidad y los servicios ecosistémicos*. https://doi.org/10.62486/latia202587

Kayusi, F., Kasulla, S., & Malik, S. J. (2024). *Climate Information Services (CIS): A Vital Tool for Africa ' s Climate Resilience*. *18*(10), 108–117.

Madhumita, G., Diana, P. D., Kiran, P. B. N., Aggarwal, S., & Nargunde, A. S. (2024). AI-powered Performance Management: Driving Employee Success and Organizational Growth. *2024 5th International Conference on Recent Trends in Computer Science and Technology (ICRTCST)*, 204–209.

Madhumithaa, N., Sharma, A., Adabala, S. K., Siddiqui, S., & Kothinti, R. R. (2025). Leveraging AI for personalized employee development: A new era in human resource management. *Advances in Consumer Research*, *2*, 134–141.

Malik, A., Budhwar, P., Mohan, H., & NR, S. (2023). Employee experience–the missing link for engaging employees: Insights from an MNE's AI-based HR ecosystem. *Human Resource Management*, *62*(1), 97–115.

Manoharan, P. (2024). A Review on AI Driven HR Systems: Revolutionizing HR Systems and Talent Management. *Sch J Eng Tech*, *6*, 179–184.

Mer, A., & Srivastava, A. (2023). Employee engagement in the new normal: Artificial intelligence as a buzzword or a game changer? In *The adoption and effect of artificial intelligence on human resources management, Part A* (pp. 15–46). Emerald Publishing Limited.

Mittal, P., Jora, R. B., Sodhi, K. K., & Saxena, P. (2023). A review of the role of artificial intelligence in employee engagement. *2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS)*, *1*, 2502–2506.

Nusair, R. E. (2025). The Impact of Artificial Intelligence on Recruitment and Employee Engagement in Human Resources. *International Journal of Academic Publishing in Educational Sciences and Humanities (IJAPESH)*, 30–42.

Nyathani, R. (n.d.). *Revolutionizing HR: AI-Driven Digital Transformation for a Competitive Edge*.

Nyathani, R. (2023). AI in performance management: Redefining performance appraisals in the digital age. *J. Artif. In Tell. Cloud Comput. SRCIJAICC-146*, *134*, 2–5.

Okatta, C. G., Ajayi, F. A., & Olawale, O. (2024). Navigating the future: integrating AI and machine learning in hr practices for a digital workforce. *Computer Science & IT Research Journal*, *5*(4), 1008–1030.

Parasa, S. K. (2022). Impact of AI on Employee Onboarding in HR Transformation. *Available at SSRN*, *5102766*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5102766

Tariq, M. U. (2024). AI and the future of talent management: Transforming recruitment and retention with machine learning. In *Global practices on effective talent acquisition and retention* (pp. 1–16). IGI Global.

Yanamala, K. K. R. (2024). Artificial Intelligence in talent development for proactive retention strategies. *Journal of Advanced Computing Systems*, *4*(8), 13–21. https://doi.org/10.69987/JACS.2024.40804

CONFLICT OF INTEREST

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AUTHOR'S RESPONSIBILITY STATEMENT

Fredrick Kayusi, Petros Chavula, Michael Keari Omwenga: conceptualization, methodology, formal analysis, development of results, writing – initial draft.

Linety Juma, Bismark Agura Kayus: methodology, formal analysis, development of results, visualization, writing – review and editing.

Bismark Agura Kayus , Rashmi Mishra: supervision, validation, review of the final document.